Revision 2026 • Semester II

Fundamentals of Electrical Engineering Lab

Course Code: 2039 | Credits: 1.5 | Contact Hours: 45 | Lab Course

A practical journey into the core principles of electrical engineering, from verifying Ohm's Law to mastering complex network theorems and AC power analysis.

Download Lab Manual / Notes

Official Source Declaration

This digital lesson is architected based on the **Official Syllabus for Course 2039 (Revision 2026)** provided by the State Institute of Technical Teachers' Training & Research (SITTTR), Kerala.

- **Authoritative Source:** SITTTR Syllabus PDF (daefd34c...).
- **Scope:** All 4 Modules, 12 prescribed experiments, and the open-ended experiment framework are included.
- **Accuracy:** Technical values, CIE/ESE marks, and Course Outcomes are mapped directly from the official document.

Course Dashboard

Teaching Scheme

L:T:P:R — 0:0:3:0

Total Hours: 45 Hours

Self-Learning: 22.5 Hours

Assessment

CIE: 60 Marks

ESE: 40 Marks

Total: 100 Marks

Pre-requisites

Course: 1031

Title: Basics of Electrical & Electronics Engineering

How to Study this Lab Course

This lab is designed to bridge the gap between theory and practice. Follow these steps for each experiment:

1. **Pre-Lab:** Study the relevant theorem (e.g., Thevenin's) in your theory textbook.
2. **Simulation:** Use the virtual lab links provided in the references to simulate the circuit before physical assembly.
3. **Execution:** Connect the circuit on the workbench, ensuring all meters (Ammeter, Voltmeter) are in the correct polarity.
4. **Analysis:** Compare experimental results with theoretical calculations. If there is a deviation, identify the cause (e.g., internal resistance of meters).

Course Outcomes (CO)

CO#	Course Outcome Description	Bloom's Level
CO1	Examine Ohm’s law through laboratory experiments and explain the voltage–current relationship.	Apply
CO2	Examine basic circuit laws (KCL, KVL, voltage division, and current division) through practical experiments.	Apply
CO3	Examine basic electrical network theorems.	Apply
CO4	Determine the power and power factor in electric circuits using appropriate measuring instruments.	Apply

Syllabus Coverage Map

Module-wise Experiment Distribution

Module	Focus Area	Experiments	Hours
Module I	Ohm's Law & Resistance	Expt 1, 2, 3	9
Module II	Kirchhoff's Laws & Dividers	Expt 4, 5, 6, 7	12
Module III	Network Theorems	Expt 8, 9, 10	9
Module IV	Power & Power Factor	Expt 11, 12 + Open Ended	15

Learning Roadmap

Step 1: The Foundation (Module I) - Understand the linear relationship between V and I. Learn how temperature affects resistance.

Step 2: Circuit Analysis (Module II) - Master the laws that govern every node and loop in a circuit. Learn how to predict voltage and current without complex math using division rules.

Step 3: Advanced Simplification (Module III) - Learn to simplify complex networks into a single source and resistor (Thevenin) and find the condition for maximum energy transfer.

Step 4: Real-world Measurement (Module IV) - Move from DC to AC. Learn to measure not just "apparent" power but "true" power and understand the efficiency (Power Factor) of a circuit.

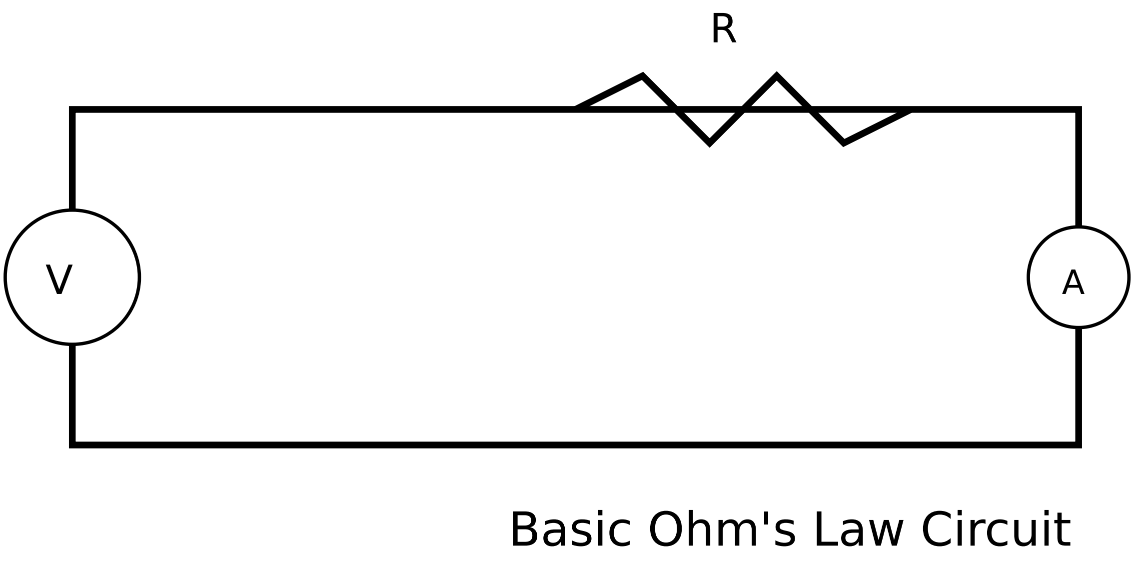
Module I: Ohm's Law & Resistance

9 Hours | CO1

Experiment 1: Verification of Ohm's Law

Objective: To determine the voltage-current relationship in a linear circuit.

Ohm's Law states that at constant temperature, the current (I) flowing through a conductor is directly proportional to the potential difference (V) across its ends.



Circuit for Ohm's Law Verification

Malayalam support: Ohm's Law □□□□□□□ □□□ □□□□□□□□□□□□ □□□□□□□□□ □□□□□□□□□ □□□□□□□ □□□□□□□□□ ($V \propto I$). □□□□□□□ □□□□□□ □□□ □□□□□□ (Straight line) □□□□□□□□□□□.

Ohm's Law Calculator

Voltage (V)

Enter Volts

Resistance (Ω)

Enter Ohms

Result: I = 0 A

Calculate Current

Experiment 2 & 3: Resistance and Temperature

Hot & Cold Resistance: An incandescent lamp's filament is made of tungsten. Its resistance increases significantly as it heats up. Cold resistance is measured using a multimeter when the lamp is OFF. Hot resistance is calculated using V/I when the lamp is ON.

Example Calculation:

Given:

Lamp rated 230V, 60W. Measured Current (I) = 0.26A at 230V.

Required:

Hot Resistance (R_h).

Calculation:

R = V / I = 230 / 0.26 = 884.6 Ω.

Answer:

Hot Resistance is 884.6 Ω.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

Module II: Kirchhoff's Laws & Dividers

12 Hours | CO2

Experiment 4 & 5: KCL and KVL

- **KCL (Kirchhoff's Current Law):** The algebraic sum of currents entering a node is zero. ($\Sigma I = 0$).
- **KVL (Kirchhoff's Voltage Law):** The algebraic sum of all voltages around any closed loop is zero. ($\Sigma V = 0$).

Malayalam support: KCL [illegible]. KVL [illegible].

Experiment 6 & 7: Division Rules

Voltage Division (Series)

$V_x = V_{total} * (R_x / R_{total})$

Current Division (Parallel)

$I_1 = I_{total} * (R_2 / (R_1 + R_2))$

Module III: Network Theorems

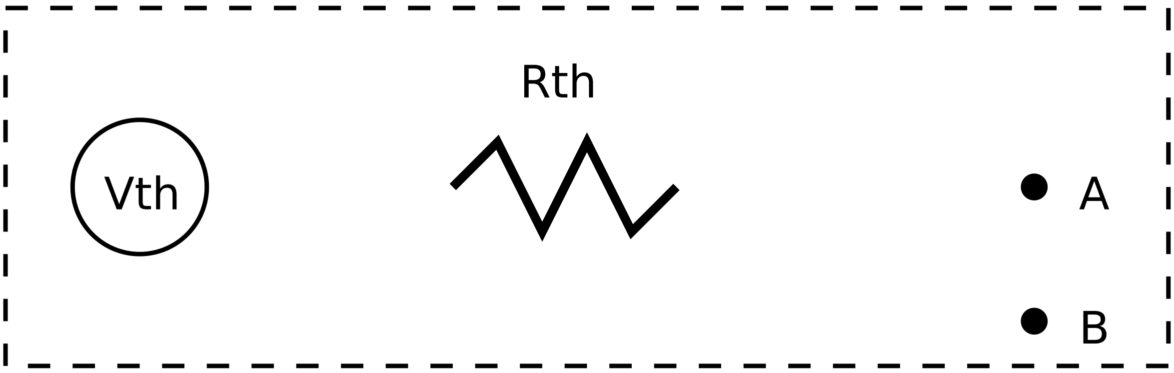
9 Hours | CO3

Experiment 8: Superposition Theorem

In a linear bilateral network containing multiple sources, the current through any branch is the algebraic sum of currents produced by each source acting alone, while other sources are replaced by their internal resistances.

Experiment 10: Thevenin's Theorem

Any linear circuit can be simplified to an equivalent circuit with one voltage source (V_{th}) and one series resistance (R_{th}).



Thevenin Equivalent Circuit

Module IV: Power & Power Factor

15 Hours | CO4

Experiment 11: DC Power Measurement

Power in DC is measured using a Wattmeter. It can be verified using Voltmeter and Ammeter readings ($P = V \times I$).

Experiment 12: AC Power Factor

In AC circuits, Power Factor ($\cos \varphi$) = True Power (W) / Apparent Power (VA).

> **Malayalam support:** AC [illegible] [illegible] [illegible] [illegible] [illegible] Power Factor [illegible]. [illegible] 1 [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

Open-ended Experiment

Objective: To encourage creativity and application of knowledge.

Students must design their own experiment. Examples from syllabus:

- Examine Norton Theorem and note the inference.
- Measurement of power and power factor of a LED tube light.

Student Activities for Self-Learning

Week	Activity Description	Hours
1-3	Study symbols, meter connections, and temperature coefficients.	4.5
4-7	Virtual lab simulations for KCL/KVL and division rules.	6.0
8-10	Review Network Theorems and complete virtual lab post-tests.	4.5
11-15	Prepare proposal for Open-ended experiment and document results.	7.5

Formula Bank

Ohm's Law

$V = I \times R$ (Volts)
$I = V / R$ (Amperes)
$R = V / I$ (Ohms)

Power Formulas

$P = V \times I$ (DC Watts)
$P = V \times I \times \cos\varphi$ (AC True Power)
$\cos\varphi = P / (V \times I)$

Visual Library

[Ohm's Law Circuit]
[Thevenin Equivalent]
More diagrams in Lab Manual

Assessment Methodology

Component	Method	Marks	Schedule
Attendance	Continuous	5	End of Sem
Continuous Eval (CE)	Lab Performance	30	Every Slot
Practical Test 1 (CA1)	Internal Exam	15	7th Week
Practical Test 2 (CA2)	Internal Exam	15	14th Week
Open Ended (OEE)	Project/Experiment	10	15th Week
Total CIE		**60**	
ESE	Final Practical	**40**	End of Sem

Module-wise Questions (Q&A)

Q1: What is the difference between Hot and Cold resistance? (Module I)

Cold resistance is the resistance of a component (like a lamp filament) at room temperature. Hot resistance is the resistance when the component is operating and heated up. For metals like tungsten, Hot resistance is much higher than Cold resistance.

Q2: State KCL and its significance. (Module II)

Kirchhoff's Current Law states that the total current entering a junction equals the total current leaving it. It is based on the Law of Conservation of Charge.

Q3: When is Maximum Power Transfer achieved? (Module III)

Maximum power is transferred from a source to a load when the load resistance (R_L) is equal to the internal resistance or Thevenin resistance (R_{th}) of the source.

Q4: Why is Power Factor important in AC circuits? (Module IV)

A low power factor means more current is required to perform the same amount of useful work, leading to higher energy losses in the wires and equipment.

Practice Model Paper

Note: This is a practice paper, not an official examination paper.

Part A: Write-up (10 Marks)

1. Write the aim, circuit diagram, and procedure for verifying Thevenin's Theorem.

Part B: Execution (10 Marks)

2. Rig up the circuit for Voltage Division Rule and measure voltages across three series resistors.

Part C: Results & Viva (20 Marks)

3. Tabulate readings, calculate errors, and answer viva questions on Power Factor.

Quick Revision

- **Ohm's Law:** V=IR. Valid for linear conductors at constant temperature.
- **KCL:** Junction Law (Conservation of Charge).
- **KVL:** Loop Law (Conservation of Energy).
- **Thevenin:** Simplifies circuit to V_{th} and R_{th}.
- **Power Factor:** cos φ = W / (V × I). Range is 0 to 1.

> **Exam Tip:** [illegible]

Glossary

Linear Circuit
: A circuit where parameters like resistance do not change with voltage or current.

Node
: A point in a circuit where two or more circuit elements meet.

Wattmeter
: An instrument used to measure electrical active power in watts.

Multimeter
: A versatile tool that can measure voltage, current, and resistance.

References

- A Textbook of Electrical Technology- Volume I & II, B. L Theraja, S Chand.
-
- IIT Roorkee Virtual

Search Handbook